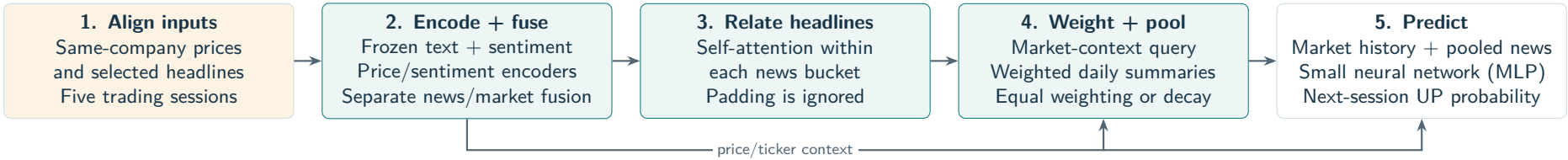
A working testbed; useful news forecasts remain unproven

Research focus: does news retain forecasting value as it gets older, and does explicit age weighting help?

Progress: built and audited a public company-news pipeline, then completed two controlled temporal comparisons. **Neither established a meaningful forecasting benefit.**

Since the first brief: NIFTY-news/SPY prototype → StockNet and SPNews adaptations → current CMIN-US company panel. Earlier experiments and negative results are preserved. This is an independent adaptation, not an empirical reproduction of FININ.

Current pipeline: the pooled version used in the recent tests



Before the final MLP: five market vectors (80 values) + one pooled news vector (16 values) = **96 inputs**. Fused prices/ticker context supplies the market query and also feeds the predictor. Sentiment joins each headline earlier; weights shape the summary rather than becoming extra columns. The initial baseline used 160 concatenated inputs; these tests match the pooled geometry.

Inputs, in plain language

News: company-linked headline titles; at most six per arrival session, up to 30 across the history. Frozen sentiment gives positive, neutral and negative scores.

Market: six price/volume features per session, plus a fixed company-ticker text description. No future prices enter the inputs.

Frozen: RoBERTa text and finance TinyBERT sentiment. **Trained:** projection, numeric/fusion layers, attention and predictor.

Evidence available

12 companies; 9,984 eligible company-days.

2,004 evaluated cases on 167 market dates.

Two chronological evaluation periods: May–August 2020 and January–April 2021; two random initializations per model.

Coverage is uneven; one company has no development news. Company-days on the same market date are not independent.

One forecast: what is known, and when?

News enters its first eligible exchange-close bucket. In an ordinary trading week: Monday source date → Wednesday forecast → Thursday answer. This delay is a timing assumption, **not verified historical availability**; the relevant information may already be priced in.

What was tested, and what can we report?

Test A / Does explicit decay help?

Keep the same news, five-session price history and model geometry. Compare equal temporal weighting with a **48-hour half-life**: each extra 48 hours halves a story's multiplier, before normalization.

Result: a tiny loss reduction of **0.000324**; the uncertainty interval includes both improvement and worsening. No practical benefit was established.

Eight fits. The age-removal curve was tiny and non-monotone; it did not identify a market decay law.

Test B / Does older news add value?

Keep five price sessions in every model. Compare prices only, prices + the **current arrival bucket**, and prices + current and four older buckets.

Result: adding older news **increased loss by 0.000560** on average. Seeds disagreed; no useful older-news addition was established.

Eight new fits; four matching all-news fits reused. "Current" means proxy arrival, not verified fresh publication.

Scores on the same evaluation dates

Model / reference	Log loss	How to read the result
Constant training-prevalence probabilities	0.689820	Reference using no daily news or price changes
Prices only (matched pooled model)	0.689771	Almost the same as the constant reference
Prices + current news bucket	0.689651	Lowest point estimate; advantage is tiny and uncertain
Prices + current and older news, equal weighting	0.690210	Older-news addition did not improve forecasting
Prices + current and older news, 48-hour decay	0.689887	Tiny change; no reliable decay benefit

Log loss measures probability quality; lower is better. Results average model losses across two initializations, not an ensemble. The predeclared practical improvement was 0.005; neither temporal test met it.

What is verified

The implementation consumes news: changing inputs can change predicted probabilities. Saved forecasts, timing rules, model states and comparisons were independently audited.

Changing a prediction does not mean improving it. Forecasts remained mostly UP; both later-period decay models predicted UP throughout.

What remains unresolved

Exact publication/availability times and some source-date conflicts. Whether weak results come from timing, relevance, representations, learning or weak predictability. These periods were already inspected: the results are **exploratory**. The experiments do not establish causality, trading value or information diffusion.

Meeting takeaway: "I tested explicit decay and older-news contribution using matched comparisons. Neither delivered a meaningful forecasting gain. I preserved the negative results and stopped further model variations."

Decision for the next stage: agree the data/timing standard and empirical question before another model. A timestamped earnings-release/movement-size approach was discussed only; it has **not been built or validated**.

Evidence: `reports/cmin_news_decay_2026-10-14.md`; `reports/cmin_older_news_2026-10-14.md`; their saved experiment artifacts and independent final reviews. Conditional 95% loss-difference intervals: decay minus equal weighting $[-0.001176, +0.000539]$; all-news minus current news $[-0.000031, +0.001155]$. Date-block intervals retain companies together and omit training uncertainty.

Reference: Wang, Cohen & Ma (2024), *Modeling News Interactions and Influence for Financial Market Prediction*, Sections 4.2–4.3 and 6.4. FININ has no explicit exponential decay term; this public-data study is an independent extension.